RAJIV GANDHI INSTITUTE OF PETROLEUM TECHNOLOGY

Digital Image Processing

Mid-Semester Examination

B.Tech CSE,IT,CSD, Minor & IDD: VII Semester

Time: 120 Minutes | Maximum Marks: 31

GENERAL INSTRUCTIONS

- All questions are compulsory.
- Section A: show essential working and write every requoired output image matrix clearly.
- Section B: choose one best answer from Annexure I. Only 15 of the 45 alternatives are used.
- Unless stated otherwise, matrix entries are grayscale pixel intensities.
- Use the application context to identify the image-processing operation; keyword matching alone may be misleading.

SECTION A | Applied Image-Processing Problems (4 x 4 = 16 marks)

Q1. CCTV / Industrial Camera Contrast Enhancement - 4 marks

A machine-vision camera inspects identification numbers printed on metallic components moving on a production line. Poor illumination compresses the useful gray levels into a limited range. One captured 8-bit image patch is:

50	60	70	80
90	100	110	120
130	140	150	160
170	180	190	200

The engineer decides to perform linear contrast stretching so that input intensity 50 becomes 0, input intensity 200 becomes 255, and all intermediate intensities are mapped linearly between them.

(a) In this inspection system; explain what contrast stretching does and why it may improve machine or human interpretation. **[1]**

(b) Apply the specified stretch to every pixel. Write the complete 4 x 4 enhanced image matrix. Round to the nearest integer where necessary. **[2]**

(c) Comment on what happens to the separation between originally close gray levels. Does this operation create new physical detail that was absent in the captured image? **[1]**

Q2. Medical / Low-Light Imaging: Histogram Equalization - 4 marks

A compact imaging device produces the following low-contrast 3-bit grayscale image. The available gray levels are 0 to 7:

0	0	1	1
0	0	1	1
2	2	3	3
2	2	3	3

The engineer applies histogram equalization. For this question, first count the pixels at each input gray level, convert the counts to proportions, form the cumulative proportion up to each level, map it onto the available output range 0-7, and take the integer part of each mapped level.

(a) Explain the purpose of histogram equalization in an imaging application. **[1]**

(b) For input levels 0, 1, 2 and 3, determine the histogram counts, cumulative proportions and the corresponding new output gray levels. **[1.5]**

(c) Replace every original pixel by its mapped value and write the complete 4 x 4 equalized image matrix. **[1]**

(d) State whether histogram equalization must always produce a perfectly uniform histogram in a small discrete digital image. **[0.5]**

$S_k = S_{min} + \frac{S_{max} - S_{min}}{r_{max} - r_{min}} (r_k - r_{min})$

$I = \frac{I_0 - I_{min}}{I_{max} - I_{min}}$

Q3. Automated Defect / Boundary Detection Without a Direct High-Pass Kernel - 4 marks

A visual quality-control system examines a manufactured sheet. Two adjoining areas have very different brightness values:

10	10	100	100
10	10	100	100
10	10	100	100
10	10	100	100

The company wants a simple two-stage procedure to locate the transition without directly using Sobel, Prewitt, Laplacian or another high-pass kernel.

Stage 1 - slowly varying reference: For each row, keep the first pixel unchanged. From the second pixel onward, replace each position by the average of the current pixel and the pixel immediately to its left. Call the resulting image L.

Stage 2 - local change: Find the absolute difference between the original image and L. Call this D. In the final binary edge image E, write 1 wherever D is 20 or greater, and 0 elsewhere.

(a) Explain intuitively what low-frequency and high-frequency regions mean in an image, relating them to nearly uniform surfaces and abrupt transitions. **[1]**

(b) Calculate and write the complete matrices L (slowly varying reference), D (absolute difference), and E (binary edge image). **[2.5]**

(c) Identify the detected vertical transition and briefly explain why the local difference becomes large there. **[0.5]**

Q4. Drone / Edge-AI Image Transmission: Spatial Reduction Before Compression - 4 marks

A drone inspects pipelines and sends images through a bandwidth-limited wireless link. Before encoding, an edge-processing unit reduces spatial size to lower the amount of image data. Consider:

10	20	30	40
30	40	50	60
50	60	70	80
70	80	90	100

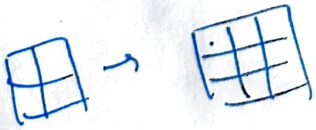
Reduce the image from 4 x 4 to 2 x 2. For each output pixel, use bilinear interpolation at the centre of the corresponding 2 x 2 input neighbourhood.

(a) Explain image compression in the context of transmitting drone or mobile-camera imagery. State one advantage and one possible disadvantage. **[1.5]**

(b) Determine the four output pixel values and write the complete 2 x 2 reduced image. **[1.5]**

(c) The original image has 16 pixels and the reduced image has 4 pixels. State the pixel-count reduction ratio. **[0.5]**

(d) Comment on why spatial reduction can lower data requirements but may reduce fine spatial detail. **[0.5]**



SECTION B | Modern Image Processing / AI Applications (15 x 1 = 15 marks)

Direction: For each situation, select one best answer from Annexure I and write the exact term/value in the blank. The distractors are intentionally related; use the stated operation and application context. **WRITE QNo____(fill in the blank)**

B1. PCB / Electronics Inspection

A PCB inspection camera records a patch in which intensity changes sharply from dark substrate to a bright metallic conductor. Before any object segmentation is performed, this local evidence of rapid spatial intensity change is an _____.

B2. Boundary of a Segmented Object

A smart camera has already segmented a person from the background. The connected contour separating the foreground object region from surrounding background pixels is the object's _____.

B3. Industrial Low-Contrast Image Enhancement

A machine-vision camera sees a metallic component whose useful intensities lie mainly between 80 and 130. The software uses a linear point-wise mapping: the lowest useful value is mapped to black, the highest to white, and intermediate values proportionally. No histogram/CDF statistics are used. The operation is _____.

B4. Fixed-Camera Traffic Monitoring

A fixed traffic camera maintains a reference image/model of the empty road. Each current frame is compared with that stationary reference, and sufficiently changed pixels are labelled as possible foreground objects. The classical procedure is _____.

B5. Creating Descriptors for an AI Classifier

After isolating an industrial object, the system replaces its raw pixels by a descriptor such as [area, perimeter, centroid, circularity, corner count] for subsequent recognition. Producing these informative measurements is _____.

B6. Detecting Long Pipeline / Road Features

A drone image contains separated edge pixels from a long pipeline. Individual edge pixels contribute evidence for possible straight-line parameters, and a strong accumulation of compatible votes indicates a likely line. This technique is the _____.

B7. Classical Filtering and CNN Foundations

A 3 x 3 numerical filter is rotated by 180 degrees before it is slid across an image. At each location, corresponding neighbourhood and filter values are multiplied and summed. This operation is _____.

B8. CNN Visual Representation

After a learned CNN filter and nonlinear activation are applied across an image, a 2-D activation array shows where the learned visual pattern responds strongly. This array is a _____.

B9. CNN Striding in an Edge-AI Device

A 7 x 7 feature map is processed by an ordinary contiguous 3 x 3 filter moved two pixels at a time horizontally and vertically. No border pixels are added, and an output is produced only when the complete filter remains inside the input. The output feature-map size is _____.

B10. Gradient Operator in Machine Vision

An industrial edge detector uses the kernel shown below. Compared with the corresponding Prewitt kernel, the centre row has greater weight. The kernel belongs to the _____.

-1	0	1
-2	0	2
-1	0	1

B11. Noise-Robust Edge / Blob Analysis

A scientific imaging system first performs Gaussian smoothing and then applies a Laplacian second-derivative operation to the smoothed image. The combined classical operator is the _____.

B12. Pixel-Level Scene Understanding

An autonomous-camera system labels image regions/pixels as road, vehicle, pedestrian, vegetation and background, rather than assigning one label to the whole image. The task is _____.

B13. Natural vs Man-Made Recognition

A rule-based vision system observes several long straight edges, repeated approximately 90-degree corners, pairs of approximately parallel edges, low boundary irregularity and highly regular geometry. The most appropriate class is _____.

B14. HSI Colour Processing

A camera pixel has Red = 90, Green = 60 and Blue = 30. Using the HSI Intensity definition used in the course, the intensity value is _____.

B15. Multispectral Drone Imagery

A four-band sensor contains Blue, Green, Red and NIR. A display is constructed as display Red <- NIR, display Green <- Red and display Blue <- Green. The displayed image is a _____.

ANNEXURE I - CHOICE BANK (45 alternatives; 15 are used)

Select one best choice for each Section B blank. Thirty choices will remain unused. Some alternatives include a short numerical hint; they are still only answer choices, not question formulas.

Edge	Gaussian filter	Image classification
Boundary	Gamma correction	90 - maximum of R,G,B
Corner	Contrast stretching	Hough transform
Background subtraction	Histogram equalization	Feature map
Frame differencing	Canny edge detection	True Colour Composite (TCC)
Masking	Feature extraction	Correlation
Image compression	Harris corner detection	2 x 2
Bilinear interpolation	Convolution	Natural object
Max pooling	60 - HSI intensity: mean of R,G,B	Kernel
3 x 3 - output rule: floor((N-F)/S)+1	Fully connected layer	RGB colour model
Sobel operator	Prewitt operator	False Colour Composite (FCC)
Image segmentation	5 x 5	HSI colour model
Man-made object	Laplacian operator	Centroid
LoG filter	Nearest-neighbour interpolation	Circularity
Connected-component analysis	Object detection	Stride 2

DIP Section C — Applied DIP
18 × 0.5 = 9 Marks
Fill in each blank with the one best answer from the choice bank.

1. An 8-bit, 3-band RGB image can represent a maximum of 256 different colours.
2. Which colour model used in class is preferred for contrast enhancement of a colour image while preserving colour information? HSV
3. An 8-bit grayscale image is quantized to 4 bits/pixel. It can contain 4 gray levels.
4. The .jpg image format uses the JPEG standard. JPEG stands for Joint.
5. In Canny edge detection, the operation that retains only a local maximum of gradient magnitude along the gradient direction is Non.
6. In Canny edge detection, the operation that retains a weak edge when it is connected to a strong edge is Double.
7. Which standard 3 × 3gradient operator gives weight **2** to its central row/column? Sobel
8. Gaussian smoothing followed by Laplacian edge detection is called LoG.
9. The direction-independent second-derivative operator used to emphasize rapid intensity changes is the _____ operator.
10. For averaging-based denoising, which normally gives more smoothing but also more blurring: 3 × 3or 5 × 5? _____
11. An $N \times N$image is shifted by one pixel and kept at the same size. The absolute pixel-wise difference between the shifted and original images produces a _____ image.
12. In Moore-neighbour boundary tracing, a pixel has _____ neighbouring pixels in its Moore-neighbourhood.
13. An object containing repeated straight edges, right-angle corners and regular geometry is more likely to be _____.
14. For the CNN feature-map region

$$\begin{bmatrix} 2 & 7 \\ 4 & 3 \end{bmatrix},$$

2 × 2max pooling gives _____.

15. A 7 × 7feature map is processed by a 3 × 3filter with stride 2 and no padding. The output spatial size is _____.
16. Assigning a semantic class such as road, vehicle, vegetation or background to every pixel is called _____.
17. Before convolution with an edge-detection kernel, a uint8 image is often converted to double so that _____ intermediate filter responses are not lost by integer clipping/quantization.
18. An Remote Sensing image has five bands: B, G, R, NIR and MIR. How many 3-band FCCs you can produce? _____

Rajiv Gandhi Institute of Petroleum Technology, Bangalore
B.Tech (CHE, CSE,CSD, PE,EE)
Mid-Sem Examination
Subject: HU331 Organizational Psychology
Note: Attempt all questions

Date: 22.09.26
Time 2 hrs

Full Marks: 30

1. Explain the domain of Organizational Psychology? How does Organizational Psychology contribute to the functioning of organizations? (5)

2. How do different personality types shape the quadrants in the Johari window? What does this reveal about self-awareness and team building? (5)

3. Read the case study and answer the questions that follow:

Laziness Is Contagious

Being lazy is often a quality that is shunned or looked down on in the workplace. When someone is unwilling to put energy into their work, they are, essentially, not engaged with their job. It is still unclear whether someone can have a "lazy" personality, but we can all most likely recall times when we did not want to or commit to putting forth the energy needed to do our work. In many cases, this leads to procrastination or excessive delegation, resulting in a failure to meet tight deadlines. One laziness behavior includes fleeing the scene when one does not want to work; another one is playing the victim and making excuses to make up for a lack of effort.

There is not much research on laziness, but the studies that do exist show that people think about laziness in a complicated way. People are usually comfortable describing themselves with strong positive traits, such as saying "I am very energetic." However, they rarely describe themselves as completely lazy. Instead, they use softer or diminishing words, such as "I am a little bit lazy" or "I'm sometimes lazy." People also judge laziness differently in others. A 'halo bias' means that one positive quality (such as being friendly or intelligent) can make us see someone as good in many other aspects. In contrast, a 'horn bias' means that one negative quality can make us judge a person negatively overall. Interestingly, laziness does not trigger 'horn bias' as strongly as many other negative traits. If someone is seen as a little lazy, people do not always assume they are a bad or incompetent person in every aspect of their life.

Even more worrisome, laziness can subtly escalate or catch on with others. For instance, one lazy behavior can lead to another, and sunk costs can add up to the point where you reason you will simply start over tomorrow. Recent research suggests that laziness is contagious i.e. it can spread from one person to another. The implications here are very intriguing: "For example, if your lazy boss rewards you for having invested more effort in your work, will you become

more or less lazy?". Furthermore, workers unknowingly become lazier as they rely on computer-generated responses to shape their decisions.

Regardless of the negative air surrounding laziness, some have found merit in its practice. For example, Michael Lewis, author of *Moneyball*, asserts that laziness is not necessarily a bad thing and has even helped him succeed: "My laziness serves as a filter…Something has to be really good before I'll decide to work on it."

(a) The case suggests that laziness can spread through observation and imitation. How might an organization's work structure and culture amplify this behavior and reduce overall productivity? How does the absence of time pressure influence lazy behavior? (6)

(b) To what extent is workplace laziness influenced by personality traits and motivational factors? Discuss using at least two personality theories and two motivation theories, with suitable organizational examples. (8)

(c) To what extent does relying on AI contribute to intellectual laziness among workers? Discuss with relevant examples. (6)

Rajiv Gandhi Institute of Petroleum Technology, Bengaluru

B. Tech. (PE, EE, CHE, CSE, CSD) Mid Semester Examination (Odd Sem, 2026-27)

Subject: Principles of Economics

Time: 2 hrs **Full Marks: 30**

Answer any 5 questions. Each question carries 6 marks. Use diagrams wherever possible to support and explain your answers clearly.

Question No. 1

Part A (3 marks)

"There is no such thing as a free lunch." Explain what this common expression means using Principle 1 (People face trade-offs). Suppose a student decides to spend 2 hours studying for this economics exam instead of working a part-time job that pays ₹500 per hour. What is the opportunity cost of studying?

Part B (3 marks)

Briefly explain how a business in an industry of your choice faces a similar trade-off when allocating its time or money.

Question No. 2

Part A (3 marks)

"Trade can make everyone better off." Using a simple logical argument, distinguish between Absolute Advantage and Comparative Advantage. If an experienced engineer in your industry (Petroleum, Chemicals, or Computer Science) is exceptionally skilled at *both* complex technical design and standard data entry, explain why they should still delegate data entry tasks to an assistant who is slower than them.

Part B (3 marks)

Consider two regions (Region X and Region Y) that produce two essential industrial outputs: Refined Fuel and Lines of Software Code. Assume constant opportunity costs with the following hourly production capabilities:

- Region X: Can produce either 20 units of Fuel or 20 lines of Code per hour.
- Region Y: Can produce either 10 units of Fuel or 30 lines of Code per hour.

a) Calculate the opportunity cost of producing 1 unit of Fuel for *both* regions.
b) Determine which region has the comparative advantage in Fuel and which region has it in Code.
c) Propose a mutually acceptable terms of trade (price) for 1 unit of Fuel that will benefit both regions.

Question No 3

Part A (2 marks)

State the Law of Demand in your own words. Explain the logical reason why a consumer typically buys less of an item when its price rises. Pick an industry of your choice (Petroleum, Chemicals, or Computer Science) and identify one core product. Explain how a budget-conscious buyer in that industry reacts logically to a sharp price hike for that product.

Part B (1 mark)

Explain the critical economic difference between a "change in demand" and a "change in quantity demanded.



Part C (3 marks)

Suppose the market for a specialized material or service in your industry is represented by the following simple linear supply and demand equations:

- Demand: $Q_d = 120 - 3P$
- Supply: $Q_s = 20 + 2P$

a. Calculate the equilibrium price (P) and equilibrium quantity (Q) for this market.
b. Suppose the government or an industry regulator sets a minimum price floor at ($P = 25$). Calculate the new quantity demanded and quantity supplied. State whether this creates a market surplus or a shortage, and calculate its exact size.

Question No. 4 (6 marks)

Imagine that a sudden supply disruption spikes the cost of an essential utility or raw material across your industry. Instead of letting the market adjust naturally, the government places a strict Price Ceiling (a maximum legal price restriction) at the original pre-crisis price level. Analyse the long-term consequences of this policy on consumer queues, illegal black markets, and product quality.

Question No. 5 (6 marks)

Initially, a specialized market is in equilibrium with the following functions:

- Demand: $Q_d = 100 - P$
- Supply: $Q_{s1} = 10 + 2P$

Now, a new industry regulation changes market conditions, causing the supply function to shift to: $Q_{s2} = P - 5$.

a. Calculate the initial equilibrium price and quantity.
b. Calculate the new equilibrium price and quantity after the regulatory shift.
c. State whether this regulation expanded or contracted the overall market supply, and verify your answer by calculating the change in total consumer expenditure.

Question No. 6 (6 marks)

Suppose a manufacturing plant or digital network service changes its pricing structure. At an initial price of INR 40 per unit, the quantity demanded is 120 units. When the price is raised to INR 60 per unit, the quantity demanded contracts to 80 units.

a. Calculate the Price Elasticity of Demand using preferred Midpoint Method.
b. State whether the demand is elastic, inelastic, or unit-elastic.
c. Based on your calculation, explain what will happen to the firm's total revenue following this price change.

Question No. 7

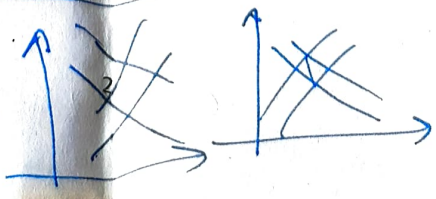
Part A (3 marks)

Define the concept of Willingness to Pay (WTP) and explain how it relates to Consumer Surplus as described in Mankiw's text. Suppose a business in your field (Petroleum, Chemicals, or Computer Science) launches a brand-new software license or refined input material. If a buyer's willingness to pay for this asset is ₹5,000, but the market price is set at ₹3,500, explain logically what the buyer chooses to do and calculate their resulting individual consumer surplus.

Part B (3 marks)

State the definition of an efficient allocation of resources. Analyse how a perfectly competitive, unregulated market automatically maximizes Total Surplus without needing a central planner. In your analysis, explain what happens to economic efficiency if a firm produces *fewer* units or *more* units than the natural equilibrium quantity.

$100 - P = P - 5$
$105 = 2P$
$2P = 95$
$P = 47.5$





Roll No. _____

RAJIV GANDHI INSTITUTE OF PETROLEUM TECHNOLOGY

(An Institute of National Importance)

Mid-Semester Examination, Odd Semester, 2025-2026

B. Tech. 4th YEAR (CSE+CSD+IT+IDD)

Course Name : Generative Artificial Intelligence | **Time: 2 Hours**

Course Code : CS441 | **Max. Marks: 30**

Instructions:

1. Do not write anything other than your roll number on the question paper.
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

1. **(3 Marks)** Consider a GAN consisting of a generator G and a discriminator D, where the generator takes a latent vector z as input and produces a generated sample G(z).

A. Derive the standard GAN minimax objective function step by step. Explain the role of each term in the objective.

B. Illustrate the GAN training workflow using a suitable block diagram.

2. **(4 Marks)** During one training iteration of a GAN, the discriminator D outputs the probability that each input sample is real. The outputs for three real and three generated samples are given below:

Sample	Real 1	Real 2	Real 3	Fake 1	Fake 2	Fake 3
Initial DD output	0.95	0.80	0.60	0.10	0.25	0.40
After update	0.85	0.90	0.75	0.20	0.15	0.30

Using the binary cross-entropy loss with natural logarithms:

A. Calculate the initial discriminator loss.

B. Identify the real and fake samples contributing the maximum loss.

C. Calculate the discriminator loss after the update.

D. Compare the two losses and interpret the change in discriminator performance.

3. **(3 Marks)** GAN training involves an adversarial optimization process between the generator G and discriminator D.

A. Explain any three major challenges in GAN training, with a suitable example for each.

B. Explain one suitable technique for mitigating each of the selected challenges.

4. **(3 Marks)** Consider a latent-variable generative model with joint distribution p(x,z), where x is an observed variable and z is a continuous latent variable.

A. Using Bayes' rule, derive the posterior distribution $p(z|x)$. Explain its role in inferring the latent representation of an observed sample and why its exact computation is generally intractable.

B. Using Jensen's inequality, derive a tractable lower bound on the log-likelihood $\log p(x)$ using an approximate posterior $q(z|x)$. Express the resulting ELBO in terms of the expected log-likelihood and KL divergence, and state the condition under which the bound becomes tight.

5. **(5 Marks)** Consider a latent-variable model with an observed variable x and a discrete latent variable z having four possible configurations. The joint probabilities are given by:

Latent configuration	$p(x,z_i)$
z_1	0.12
z_2	0.28
z_3	0.08
z_4	0.32

The approximate posterior distribution is given by: $q(z|x)=\{0.10,0.40,0.20,0.30\}$.

A. Calculate the marginal likelihood $p(x)$ and exact posterior probability for each latent configuration and identify the most probable configuration.

B. Using the given approximate posterior $q(z|x)$, compute the ELBO and the KL divergence $D_{KL}(q(z|x) \,//\, p(z|x))$.

C. Verify the ELBO decomposition numerically and determine whether the ELBO is tight.

6. **(2 Marks)** Consider a Variational Autoencoder (VAE) in which the encoder defines the latent distribution as $q_\phi(z|x)=N(\mu_\phi(x),\sigma_\phi^2(x))$.

A. Explain why direct sampling from the latent distribution is problematic for VAE training.

B. Derive the reparameterization trick and explain how it overcomes this limitation.

7. **(5 Marks)** For each of the following variants of deep generative models, briefly explain its architecture and working principle and relevant objective function:

A. BiGAN (Bidirectional GAN)

B. CoGAN (Coupled GAN)

C. CycleGAN

D. β-VAE

E. Ladder VAE

8. **(5 Marks)** Discuss the following evaluation strategies for deep generative models. For each strategy, explain its underlying principle, mathematical formulation, and the aspect of model quality it measures.

A. Latent-space interpolation

B. Disentanglement analysis

C. Inception Score (IS)

D. Fréchet Inception Distance (FID)

E. Precision and Recall

Rajiv Gandhi Institute of Petroleum Technology
Bengaluru Campus
Mid-Semester Examination
(ODD Semester, 2026-27)

Max. Marks: 30 **Duration:** 120 Min. **Date:** 23/9/2026 **Roll No:**
Programme/Semester: B. Tech 7th Semester **Time:** 10:00 AM-12:00 PM
Course Title: Employability Skills and Industry Readiness

S.NO	Questions	Marks
1.	During several of the industry interactions in ESIR course, the invited professionals made a similar observation: many students preferred sitting in the back rows, while only a few came forward to ask questions or take part in the discussion. One student later said: "I attended the complete session and understood what was being discussed. I did not have a question, so I did not speak. I don't think this has anything to do with my employability." (a) What do you think about this argument? Is the observation made by the industry professionals relevant to professional readiness? Give reasons. (b) At the same time, a person who speaks frequently is not necessarily a good communicator. How would you distinguish between speaking merely to get noticed and making a meaningful contribution to a professional discussion? (c) Give one example of what you would consider meaningful participation in a technical meeting or an interaction with an industry professional.	2+2+1
2.	The students listing a large number of technical and non-technical skills in their résumé. In an interview, however, simply mentioning a skill is not enough. (a) Pick any three skills from your own present profile, including at least one technical/domain skill. What can you actually do with each of these skills? (b) How would you demonstrate these skills to an interviewer rather than simply saying that you know them? (c) Suppose you are the interviewer. Give one question or task that you would use to check whether a candidate genuinely possesses any one of the skills claimed.	2+2+1
3	Companies often prefer fresh graduates who have had some exposure to industry. Consider two students from the same discipline. One has a strong academic record and sound fundamentals but has not done a formal internship. The other has an average academic record and has completed a six-month internship in a relevant industry. (a) Do six months of internship experience necessarily make the second student more employable? Give reasons for your answer. (b) If you were interviewing the student who has completed the internship what would you ask to find out whether those six months actually had a value to their prof	2+2+2

	(c) Now consider the student without an internship. What could this student show or demonstrate during the selection process to convince you that he/she is ready to work in industry?	
4.	A final year student has participated in several campus recruitment processes. The record so far is: Stage of Recruitment — No. of Companies Applied — 20 Shortlisted for written/online test — 16 Cleared the test — 9 Cleared the technical interview — 3 Offers received — 0 The student feels that there have simply not been enough opportunities and therefore plans to apply to many more companies. (a) Looking at the above record, do you agree with the student's conclusion? What does the recruitment record indicate to you? (b) Where, in your opinion, does the student need immediate improvement? Give reasons. (c) The next major placement opportunity is six weeks away. If this student approached you for advice, what would you ask him/her to work on during these six weeks? Include both technical and non-technical preparation. Also indicate how you would check whether there has actually been an improvement at the end of six weeks.	2+2+3
5.	During a technical interview, you are given a problem from your own discipline that you have not come across earlier. You understand some of the basic concepts involved, but you are unable to see the complete solution immediately. (a) How would you handle the situation? Explain how you would start and proceed with the problem. After a few minutes, the interviewer says: "I don't think your present approach is correct." (b) What would you do at this point? (c) Even if you are unable to reach the final answer, what can the interviewer learn about you from the way you handle this situation? Explain any two such attributes. (d) Think about the kind of job role for which you are currently preparing. Identify one technical area and one non-technical area in which you personally need improvement. What specific action will you take on each before your next interview?	2+1+2+2

Rajiv Gandhi Institute of Petroleum Technology, Bengaluru Campus

Mid-Semester Examination (Odd Semester, A.Y. 2026-27)

Max. Marks: 30 **Duration:** 02 Hrs. **Date:** 21/09/2026

Programme: B. Tech. CSE and CSD **Semester:** 7th **Time:** 03:00 PM - 05:00 PM

Course Title: **Sustainability and Climate Change** **Course Code:** MT 5213- 1

S.N	KL	Questions	Marks
		SECTION A	
1. Attempt ALL Parts			**(6x1)**
a)	K2	In the Brundtland formulation of sustainable development, the term *needs* gives overriding priority to the essential needs of the _____, whereas *limitations* arise from the state of _____ and social organization.	1
b)	K2	During a five-year period, an economy grows by 8%, while its total material consumption grows by 3%. Identify the type of decoupling represented by these trends. Relative	1
c)	K2	Electricity purchased from the grid for operating a data centre contributes primarily to Scope _____ emissions. fugitive	1
d)	K2	Firm X procures directly from Supplier A; A procures from B; B sources lithium from a mine. With respect to Firm X, classify B and the mine by their supply-chain tiers.	1
e)	K2	A smartphone is redesigned so that its metals and electronic components can be recovered, refurbished and returned repeatedly to production. Identify the relevant circular-economy cycle.	1
f)	K3	A climate-monitoring system must initiate protective action within milliseconds and cannot wait for a cloud round trip. Identify the most appropriate computing approach.	1
		SECTION B	
2. Attempt ALL Parts			**(2x3)**
a)	K3	A computing facility reports the following annual data: Grid electricity = 1,200 MWh; grid emission factor = 0.70 tCO_2e/MWh; diesel consumed in company-owned backup generators = 20,000 L; diesel emission factor = 2.7 $kgCO_2e/L$; purchased servers/electronic equipment = 460 tCO_2e; employee business travel = 76 tCO_2e. Classify the above activities under appropriate emission scopes and estimate the facility's Scope 1, Scope 2, Scope 3 and total annual carbon footprint. Also determine the percentage contribution of Scope 3.	3
b)	K4	A climate-resilience project today is expected to avoid environmental damage of ₹100 crore after 50 years. Determine the present value of the avoided damage at discount rates of 2% and 6%. $PV = \frac{FV}{(1+r)^t}$. Using your results, explain why the choice of discount rate has implications for intergenerational equity.	3
		SECTION C	
3. Attempt ALL Parts			**(3x6)**

a)	K4	RGIPT Bengaluru proposes replacing 500 conventional 60-W lamps with 10-W LED lamps. The lamps operate for 10 h/day and 300 days/year. Electricity costs ₹8/kWh and the retrofit requires ₹1,50,000. A 5,000 m² usable rooftop receives annual rainfall of 800 mm with a runoff coefficient of 0.80. Determine: (i) annual electrical-energy saving; (ii) annual monetary saving and simple payback period; (iii) annual harvestable rainwater; (iv) whether replacing lamps with LEDs and reducing their operating hours represent the same resource-management strategy. Justify using appropriate terminology.	1+2+2 +1
b)	K5	An electronics manufacturer reports a 50% reduction in factory waste and operates its assembly facility using renewable electricity. It therefore claims that its laptop has a sustainable supply chain. An independent third party audit, however, finds limited visibility beyond direct suppliers, uncertain origin of critical minerals, substantial use of air freight and negligible recovery of returned products. As a sustainability engineer: (i) Evaluate the validity of the company's claim by identifying three material weaknesses in its sustainability assessment. (ii) Recommend corrective interventions for any two of the identified weaknesses. The interventions should address the source of impact wherever practicable. (iii) Propose one digital technology capable of improving multi-tier supply-chain visibility. State the information problem it solves, rather than merely naming the technology.	3+2+1
c)	K6	A smart-city authority collects electricity, water, waste, traffic and air-quality data through independent systems. Sustainability reporting is presently spreadsheet-based and performed annually; consequently, abnormal resource consumption may remain undetected for extended periods. This is the same fundamental problem posed in the enclosed paper. Design a continuous sustainability-intelligence architecture for the authority. (i) Draw and explain a suitable architecture beginning with physical assets/sensors and ending with decision/action. Your architecture should incorporate at least three appropriate digital technologies covered in the course. State the function of each. (ii) Explain where edge processing and cloud processing would respectively be appropriate in your architecture. Give one sustainability-related example for each. (iii) Electricity consumption increases by 20% while occupancy and required service levels remain unchanged. Identify the first resource-management issue that should be investigated and justify your answer.	3+2+1

Bloom's Knowledge Level: K_1 – Remember K_2 – Understand K_3 – Apply K_4 – Analyze K_5 – Evaluate K_6– Create